

BAAZ JKBOSE

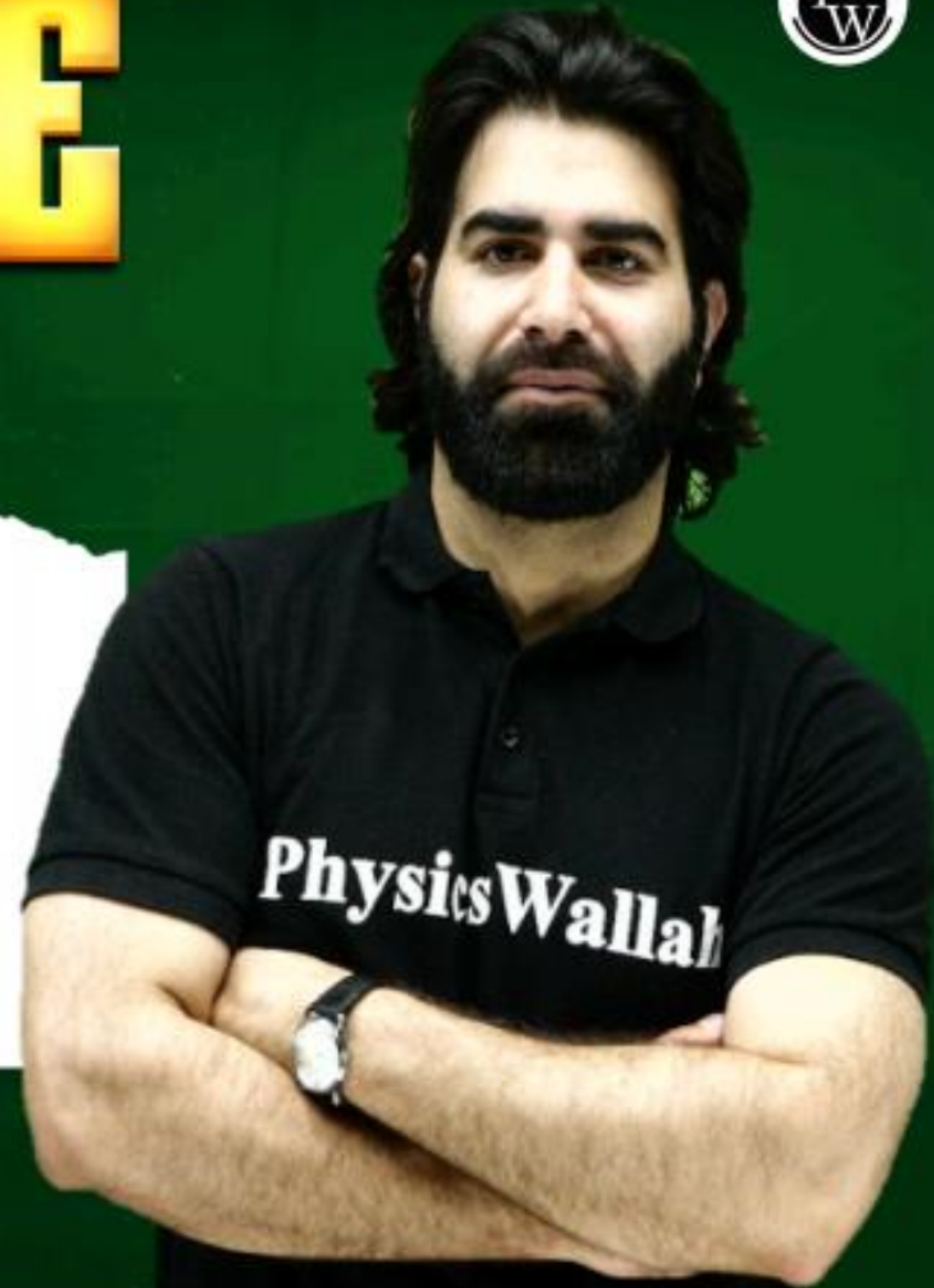
2026 CLASS-11TH

Body Fluids and Circulation

LECTURE-04

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Types of circulation, Human Heart , Lymph

Revision

→ person - Blood group AB
Antigen ① A, B
Antibody ② No

→ Erythroblastosis Foetalis → Mother ③ -
Fetus ④ +

Clotting .
Fibrinogen $\xrightarrow[\uparrow \text{Thrombin}]{\times}$ Fibrin

Lymph (NCERT = Tissue Fluid).

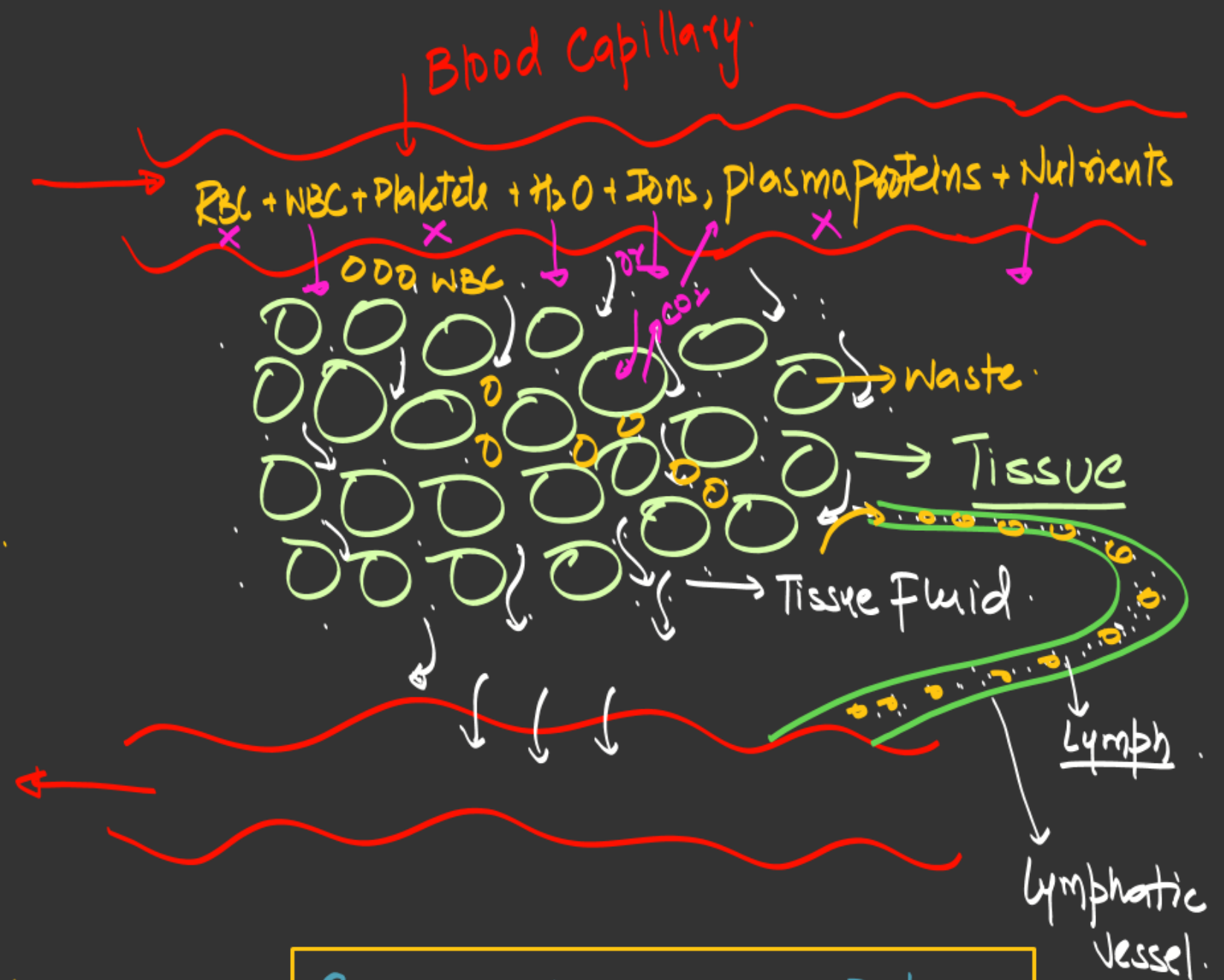
- like Blood it is Also Fluid Connective Tissue.
- It is formed from Tissue Fluid.
- It flows in lymphatic vessels.
- Finally Drains in Blood.

→ Composition lymph

Lymph = Blood - RBC - Platelets - plasma proteins

- Cells in lymph are WBC's.

- In GIT, Lymph also helps in Absorption of fat.
Giving this to (Whitish/Yellowish) colour.



Serum = Plasma - Clotting Factors
(Fibrinogen).

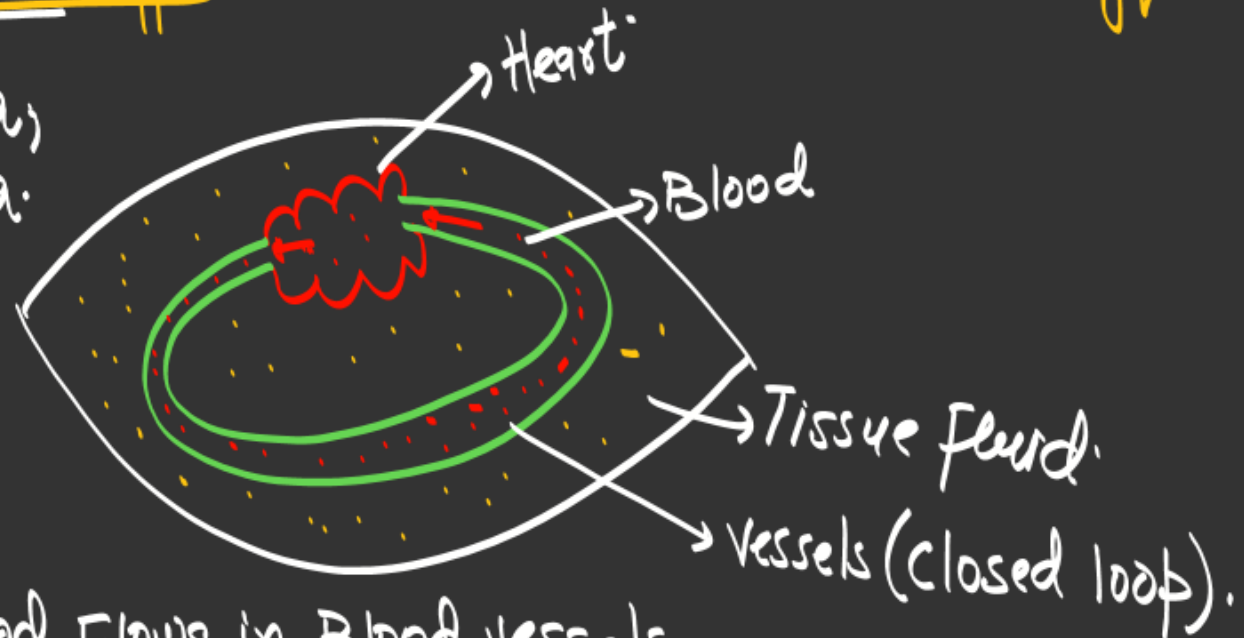
Circulatory System

- ↳ System involved in circulation.
- ↳ circulatory system helps in circulation of Nutrients, Wastes, Gases, Hormones etc.
- ↳ Basically it consists of

- Pump (Heart).
- Circulatory Fluid (like Blood).
- Vessels (Artery, vein, Capillaries) → H/W.

Closed type

e.g. Annelida, Chordata.

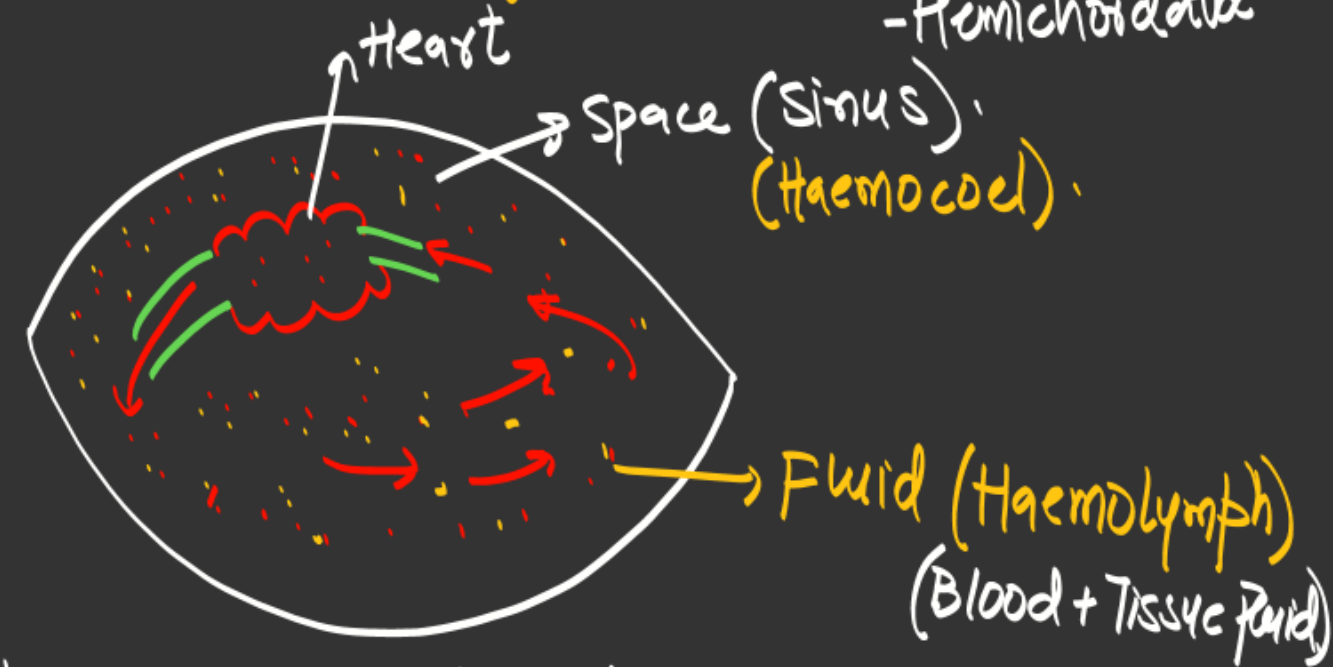


→ Blood flows in Blood vessels.

Types (Animals)

Open type

e.g. - Arthropoda
- Mollusca
- Echinodermata
- Hemichordata



→ Here Hearts pumps Blood into Body spaces (Sinuses).

Types of Circulation in vertebrate → Fishes, Amphibians, Reptiles, Aves, Mammals.

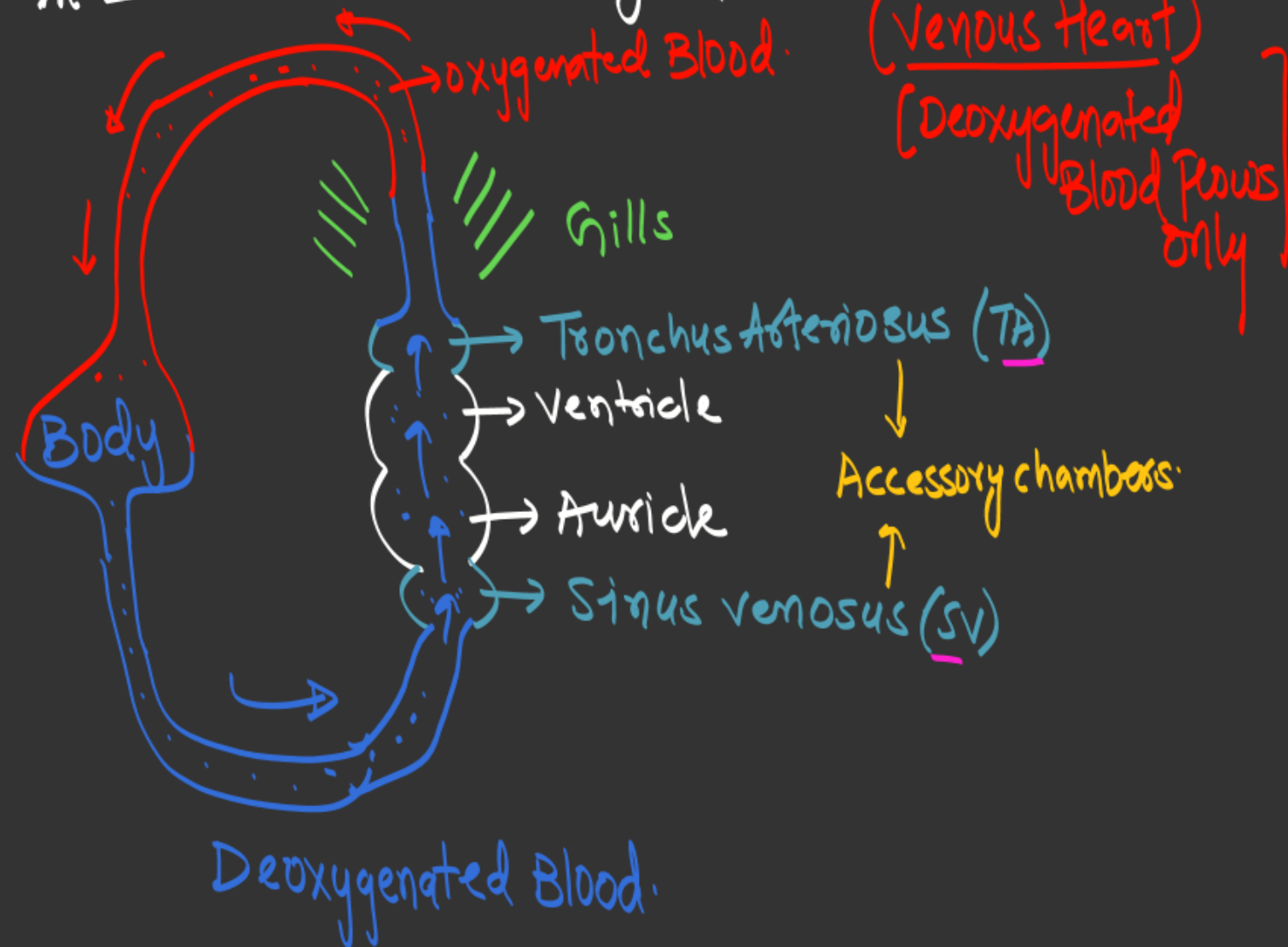
Three basic types

- ① Single circulation - Two chambered heart (Fishes)
- ② Incomplete Double Circulation (3-chambered heart - Amphibian, Reptiles)
- ③ Complete Double Circulation (4-chambered heart - Aves, Mammals).

→ Single Circulation

→ In this type of circulation Blood Flows only once in a cycle.

→ In 2-chambered heart e.g. - Fishes (Pisces)

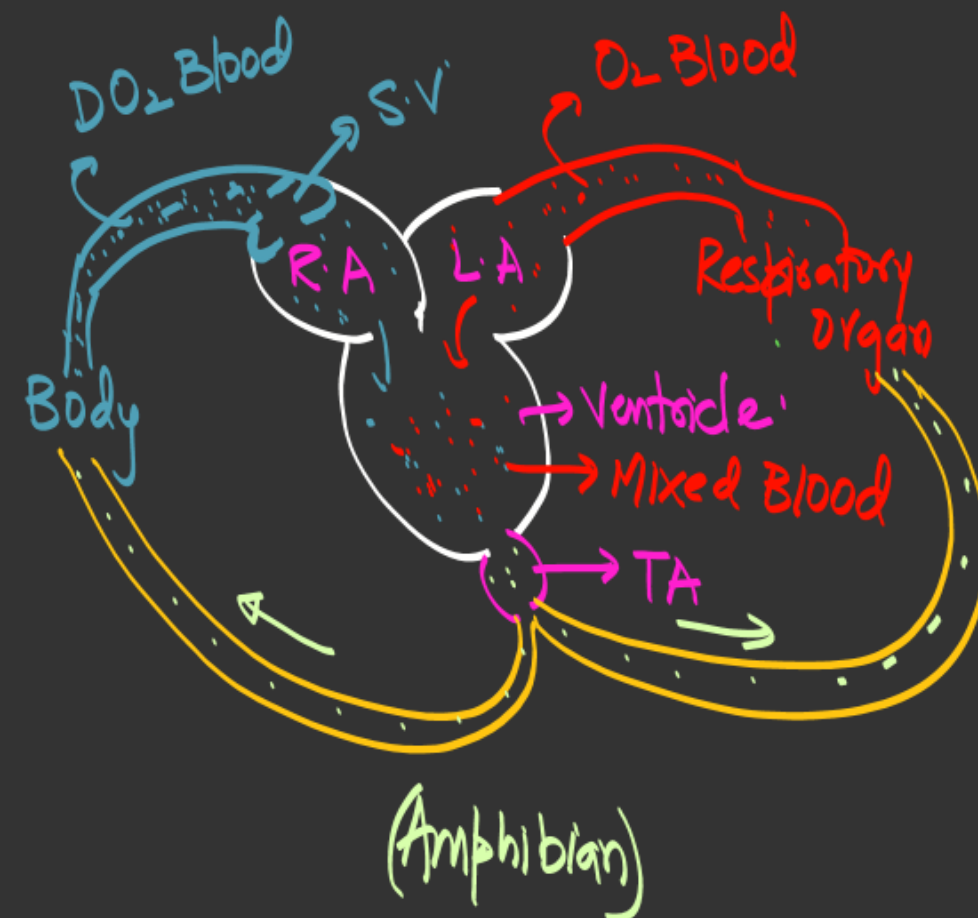


Incomplete Double Circulation

Double circulation - In one cycle the Blood flows through Heart Twice.

Incomplete - Blood gets Mixed in Ventricle.

- In 3 chambered Heart (Amphibian, Reptile).



→ No TA.

Complete Double Circulation → (Included Humans)

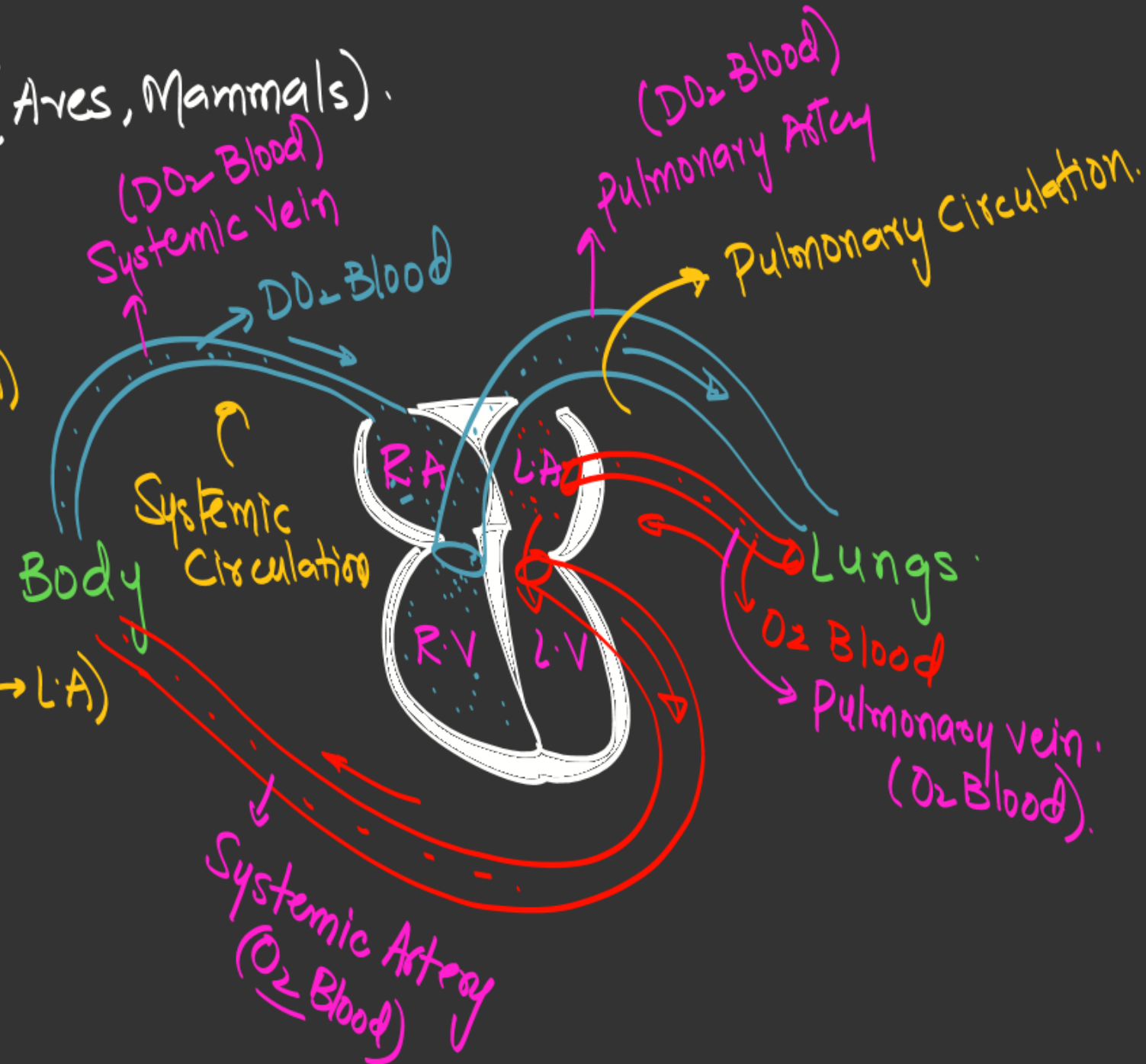
= Here Blood flows twice through Heart in one cycle and Oxygenated & Deoxygenated Blood Does not get Mixed.

⇒ 4-chambered Heart (Aves, Mammals).

⇒ Two Circulation

① Systemic Circulation:
- Btw Heart & Body (L.V → R.A)

② Pulmonary Circulation:
→ Btw Heart & Lungs (R.V → L.A)



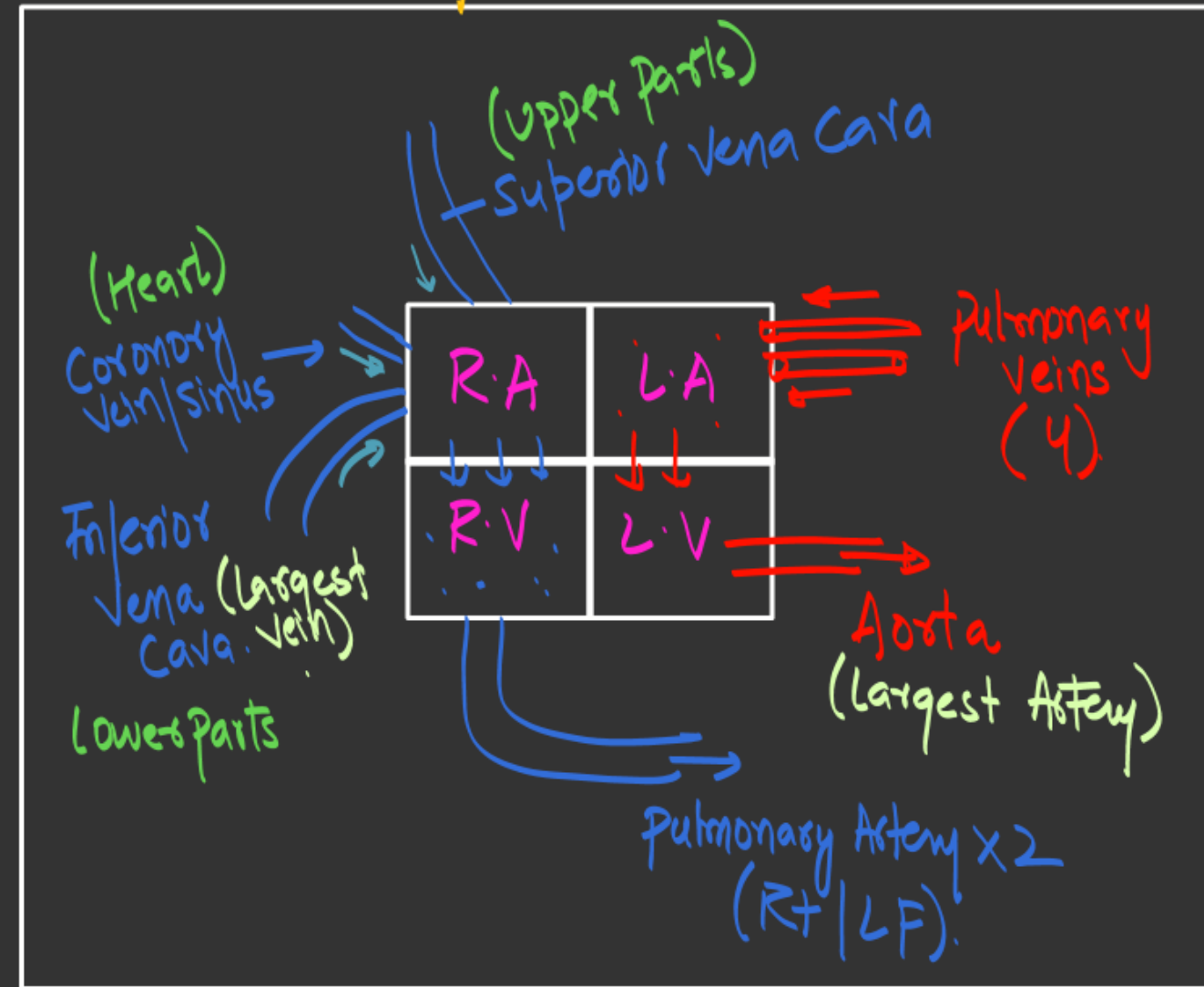
Artery = Takes Blood Away from Heart

Vein = Takes Blood towards Heart

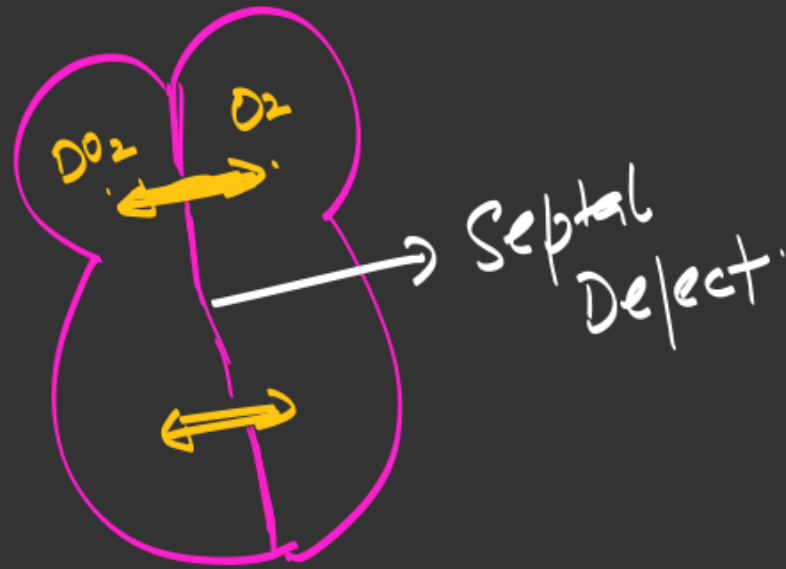
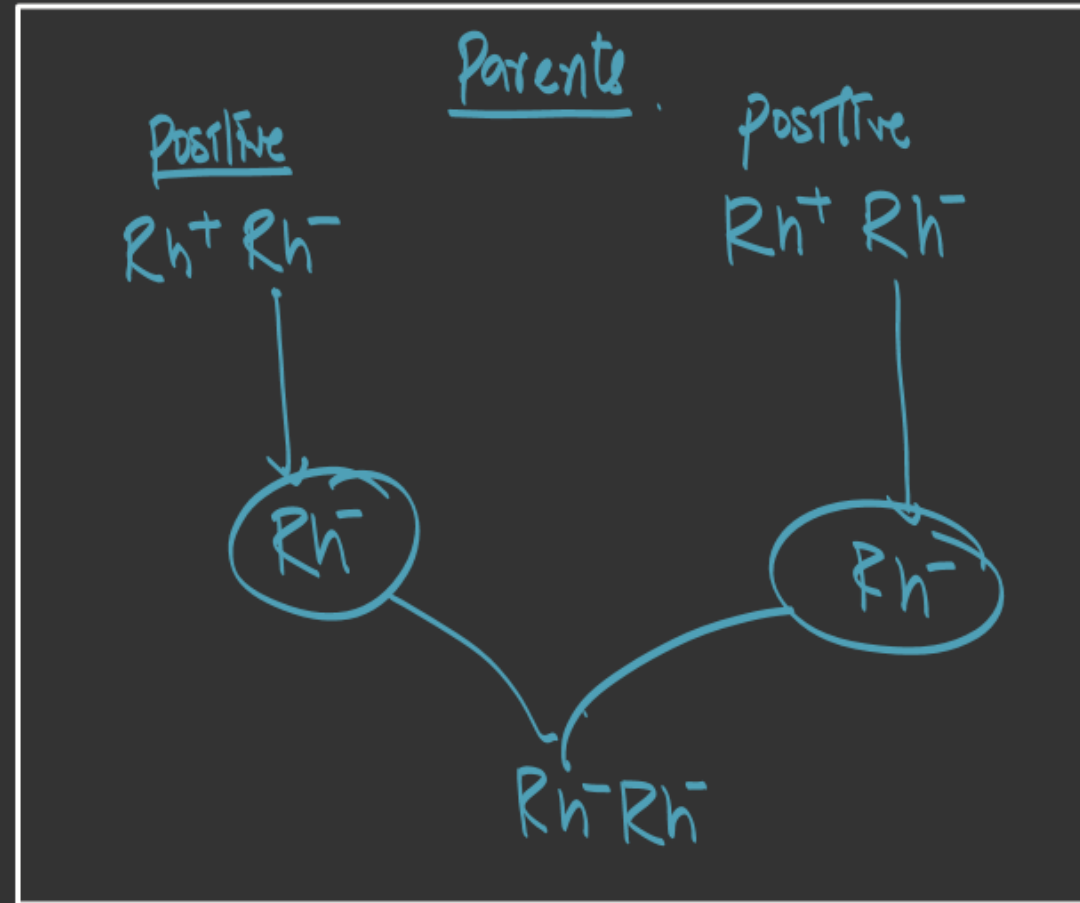
→ Human Circulation

- Closed circulatory system
- Complete Double circulation
- 4 chambered Heart

Major Blood vessels Associated with Heart



Doubt Slide



Thank
You